Quadratic Forms and the Clifford Product

From the diagonal of a bilinear form to the algebra it specifies

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Roadmap

I	One vector in both slots	what a quadratic form keeps; polarization
II	Quadratic forms at work	Mahalanobis, label propagation, Lorentz; signature
III	From a quadratic form to a product	declare Q, demand $\mathbf{v}^2 = Q(\mathbf{v})$; both halves follow
IV	The smallest algebras	one letter: complex, split-complex, dual numbers; two letters: quaternions
V	Close	what 13b and 13c build on this; exercises; reading

Sessions 10 and 11 split every bilinear form into a symmetric half and an antisymmetric half. Today we feed the same vector to both slots of a form, see which half survives, and follow that survivor all the way to the Clifford algebra.

Every number in this deck is arithmetic you can check by hand. 13b reads the algebra as a programming language with Q as its interpreter; 13c builds the wedge and the dual from it and returns to binding.

Where We Are

Recall the attention head from Session 11: two tokens $\mathbf{x}_1 = (3, 1)$ and $\mathbf{x}_2 = (1, 3)$, and a learned form $G = W_Q^\top W_K$, split into its two halves.

$$G = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}}_{S,\ \text{symmetric}} + \underbrace{\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}}_{A,\ \text{antisymmetric}}$$

	$1 \to 2$	$2 \to 1$	
$\mathbf{x}^\top S\mathbf{y}$	16	16	how alike
$\mathbf{x}^\top A\mathbf{y}$	$+8$	-8	which way

The fold of HRR keeps S and destroys A; a VSA puts order back by stipulating role atoms. Session 11 derived a product that keeps both, $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ – but only for the ordinary dot product.

Session 10 listed three sources for the matrix in the middle: *learn* it, *assume* it, or *declare* it. Geometric algebra declares it. Today: what exactly is declared.

The Question for Today

A puzzle

Declaring a geometric algebra means declaring a signature, and a signature fixes only a *symmetric* form – the half that is blind to order. Yet the algebra built from it keeps order: $\mathbf{a} \wedge \mathbf{b} = -\mathbf{b} \wedge \mathbf{a}$. Where does the other half come from?

The answer comes in three steps, one per part of the deck:

1. Put one vector into both slots of a bilinear form. What comes out is a **quadratic form**, and it sees only the symmetric half.
2. A quadratic form is the symmetric half, no more and no less: **polarization** recovers it exactly.
3. Demand a product in which every vector squares to its quadratic form. Polarization then fixes the symmetric half of the product, and the antisymmetric half comes along unconstrained, as a new kind of object.

So declaring Q is enough, because the algebra supplies the half that Q cannot see.

Part I
One Vector in Both Slots

Feed the Same Vector to Both Slots

A bilinear form takes two vectors. Give it the same vector twice:

$$B(\mathbf{x},\mathbf{y}) = \mathbf{x}^\top G\,\mathbf{y} \qquad \rightsquigarrow \qquad Q(\mathbf{x}) \;=\; B(\mathbf{x},\mathbf{x}) \;=\; \mathbf{x}^\top G\,\mathbf{x}$$

Q is the **quadratic form** of B: a function of one vector, quadratic in its coordinates. For the head:

	$G\mathbf{x}$	$Q(\mathbf{x}) = \mathbf{x}^\top(G\mathbf{x})$
$\mathbf{x}_1 = (3,1)$	$(5,\ 1)$	$3\cdot 5 + 1\cdot 1 = 16$
$\mathbf{x}_2 = (1,3)$	$(7,\ 3)$	$1\cdot 7 + 3\cdot 3 = 16$

Written as a polynomial, $Q(\mathbf{x}) = x_1^2 + 2x_1x_2 + x_2^2 = (x_1+x_2)^2$. Under this head both tokens have squared length 16.

You have met this already: $\mathbf{x}^\top\mathbf{x} = \|\mathbf{x}\|^2$ is the quadratic form of the dot product, and every squared length in the course has been a quadratic form.

The Antisymmetric Half Disappears

For any antisymmetric A, the number $\mathbf{x}^\top A\mathbf{x}$ equals its own transpose:

$$\mathbf{x}^\top A\mathbf{x} \;=\; \big(\mathbf{x}^\top A\mathbf{x}\big)^\top \;=\; \mathbf{x}^\top A^\top \mathbf{x} \;=\; -\,\mathbf{x}^\top A\mathbf{x} \qquad\Longrightarrow\qquad \mathbf{x}^\top A\mathbf{x} = 0$$

So $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x} + \mathbf{x}^\top A\mathbf{x} = \mathbf{x}^\top S\mathbf{x}$. The quadratic form sees only the symmetric half, and different forms with the same S give one and the same Q:

matrix	$Q(3,1)$	$Q(1,-1)$	$Q(2,5)$
$G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$, the head	16	0	49
$G^\top = \left(\begin{smallmatrix} 1 & 0 \\ 2 & 1 \end{smallmatrix}\right)$, the head reversed	16	0	49
$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, its symmetric half	16	0	49
$\left(\begin{smallmatrix} 1 & 7 \\ -5 & 1 \end{smallmatrix}\right) = S + 6A$	16	0	49

A head and its reverse attend in opposite directions, and their quadratic forms are identical. Whatever a quadratic form records, it is not direction.

Polarization: The Symmetric Half Comes Back

Nothing has been lost from S, though. Given only the function Q, the symmetric half can be rebuilt:

$$S(\mathbf{x}, \mathbf{y}) \;=\; \tfrac{1}{2}\big[\, Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) \,\big]$$

This is the **polarization identity**. Run it on the two tokens:

$Q(\mathbf{x}_1 + \mathbf{x}_2) = Q(4,4)$	$(4+4)^2 = 64$
$Q(\mathbf{x}_1),\;\; Q(\mathbf{x}_2)$	$16,\;\; 16$
$\frac{1}{2}\,(64 - 16 - 16)$	$\mathbf{16} \;= \mathbf{x}_1^{\top} S\, \mathbf{x}_2$

The 16 comes back. The 8 does not, and no formula in Q alone could return it: G and $G^{\top}$ have the same Q but opposite A.

So a quadratic form is a symmetric bilinear form seen along the diagonal – and the diagonal is enough to rebuild the rest of the symmetric half.

Why Polarization Works

Expand $Q(\mathbf{x}+\mathbf{y})$ with any G. The cross terms arrive in *both* orders:

$$Q(\mathbf{x}+\mathbf{y}) \;=\; \mathbf{x}^{\top}G\mathbf{x} \;+\; \underbrace{\mathbf{x}^{\top}G\mathbf{y}+\mathbf{y}^{\top}G\mathbf{x}}_{=\,2\,\mathbf{x}^{\top}S\,\mathbf{y}} \;+\; \mathbf{y}^{\top}G\mathbf{y}$$

Adding a quantity to its order-swapped copy is the half-sum of Session 11, doubled: the antisymmetric parts cancel and the symmetric parts add. With the head, $24+8=32=2\cdot 16$.

The same information, two ways

A quadratic form determines exactly one symmetric bilinear form, by polarization, and a symmetric bilinear form determines its quadratic form, by putting $\mathbf{x}$ in both slots. Declaring one is declaring the other.

The $\frac{1}{2}$ needs division by 2, which the real numbers allow; over arithmetic modulo 2 the two notions come apart, a case this course never needs.

What Counts as a Quadratic Form

A function Q on vectors is a quadratic form when (i) $Q(c\,\mathbf{x}) = c^2\,Q(\mathbf{x})$, and (ii) the polarization $Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})$ is bilinear. Equivalently: $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x}$ for some symmetric S.

function	**quadratic form?**	**why**
$x_1^2 + x_2^2$	yes	$S = I$: squared length
$(x_1 + x_2)^2$	yes	$S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$: the head
$x_1^2 + 4x_1x_2 + x_2^2$	yes	$S = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$: Session 10's indefinite form
x_1x_2	yes	$S = \left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$
$\sqrt{x_1^2 + x_2^2}$	no	scales by $\lvert c\rvert$, not c^2
$x_1^2 + x_2$	no	not homogeneous
$\lvert x_1\rvert + \lvert x_2\rvert$	no	scales by $\lvert c\rvert$; polarization not bilinear

The length is not a quadratic form; its square is. That is why squared lengths, not lengths, carry the algebra.

Reading the Matrix Off the Polynomial

Every quadratic form is a polynomial with only degree-2 terms. Its symmetric matrix is read off directly: the coefficients of the squares go on the diagonal, and half of each cross coefficient goes in each off-diagonal cell.

$$Q(\mathbf{x}) \;=\; \sum_i S_{ii}\, x_i^2 \;+\; \sum_{i<j} 2\, S_{ij}\, x_i x_j$$

polynomial	**symmetric matrix**	**a few values**
$x_1^2 + 4x_1x_2 + x_2^2$	$\left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$	$Q(1,1) = 6,\;\; Q(1,-1) = -2,\;\; Q(1, \sqrt{3} - 2) = 0$
$3x_1^2 - 2x_1x_2 + 5x_2^2$	$\left(\begin{smallmatrix} 3 & -1 \\ -1 & 5 \end{smallmatrix}\right)$	$Q(1,0) = 3,\;\; Q(1,1) = 6$
x_1x_2	$\left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$	$Q(1,1) = 1,\;\; Q(1,-1) = -1$

Many matrices produce the same polynomial – add any antisymmetric A – but exactly one of them is symmetric. That one is what a quadratic form *is*, written as a matrix.

The first row is Session 10's indefinite form: positive, negative, and zero values, all from one polynomial.

Part II
Quadratic Forms at Work

Mahalanobis: Length Measured Against the Data

Given data with mean $\boldsymbol{\mu}$ and covariance Σ, measure how far a point lies from the data with the quadratic form of Σ^{-1}:

$$Q_{\Sigma}(\mathbf{x}) \;=\; (\mathbf{x}-\boldsymbol{\mu})^{\top}\,\Sigma^{-1}\,(\mathbf{x}-\boldsymbol{\mu})$$

With $\Sigma = I$ this is squared Euclidean distance. Take instead Session 10's correlated matrix as the covariance, $\Sigma = \left(\begin{smallmatrix} 1 & 0.8 \\ 0.8 & 1 \end{smallmatrix}\right)$, so $\Sigma^{-1} \approx \left(\begin{smallmatrix} 2.78 & -2.22 \\ -2.22 & 2.78 \end{smallmatrix}\right)$:

offset from the mean	**Euclidean, squared**	**Mahalanobis, squared**
$(1,\ 1)$, along the correlation	2	1.11
$(1,\ -1)$, against it	2	10.00

Same Euclidean distance, yet the second point is three times as far (nine times, squared), because it breaks the pattern the data follows.

In NLP: Podolskiy et al. (2021) fit class means and a shared covariance to transformer embeddings of in-domain utterances, then flag out-of-domain intents by this distance, and report new state-of-the-art results on all the datasets they tested.

The Graph Laplacian: How Much a Labeling Disagrees

On a graph, the Laplacian $L = D - W$ has a quadratic form that sums squared disagreements across edges, $\mathbf{x}^\top L\,\mathbf{x} = \sum_{(i,j)\,\in\,\text{edges}} (x_i - x_j)^2$. A toy synonym chain:



labeling	good	decent	okay	mediocre	bad	$\mathbf{x}^\top L\mathbf{x}$
abrupt	1	1	1	-1	-1	4.00
smooth	1	0.5	0	-0.5	-1	1.00
constant	1	1	1	1	1	0.00

Label propagation clamps seed words (good $= +1$, bad $= -1$) and minimizes this quadratic form over the rest. The minimum is the smooth row: okay lands at 0, neutral. Rao and Ravichandran (2009) induce polarity lexicons this way, over WordNet synonym graphs.

The constant labeling costs nothing: the form measures disagreement, never level. That is why seeds are needed.

The Lorentz Form: Room for a Taxonomy

Hyperbolic embeddings of hierarchies use an *indefinite* quadratic form, one minus sign and the rest plus:

$$Q_L(\mathbf{x}) = -x_0^2 + x_1^2 + x_2^2, \qquad \mathbb{H}^2 = \{\mathbf{x} : Q_L(\mathbf{x}) = -1,\ x_0 > 0\}$$

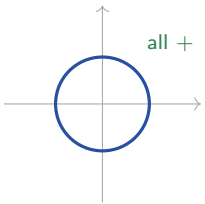
Distance on the surface is $d(\mathbf{x}, \mathbf{y}) = \operatorname{arcosh}\big(-B_L(\mathbf{x}, \mathbf{y})\big)$, where B_L is the polarization of Q_L. Worked points:

point	**coordinates**	Q_L	**distance**
$\mathbf{o}$, the origin	$(1,\ 0,\ 0)$	-1	–
$\mathbf{p}$	$(\cosh 2,\ \sinh 2,\ 0) \approx (3.76,\ 3.63,\ 0)$	-1	$d(\mathbf{o}, \mathbf{p}) = 2$
$\mathbf{q}$, the mirror image of $\mathbf{p}$	$(\cosh 2,\ -\sinh 2,\ 0)$	-1	$d(\mathbf{p}, \mathbf{q}) = 4$
a null vector	$(1,\ 1,\ 0)$	0	not on the surface

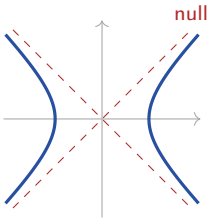
In NLP: Nickel and Kiela (2018) embed the WordNet noun hierarchy on this surface in a handful of dimensions. General concepts such as entity sit near $\mathbf{o}$, specific ones such as chihuahua far out, and the room available grows exponentially with distance, as a branching taxonomy needs.

Three Kinds of Direction

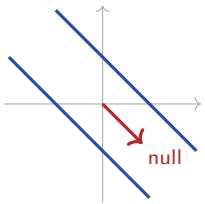
Draw the set where $Q(\mathbf{x}) = 1$. Its shape *is* the quadratic form:



$x_1^2 + x_2^2$: a circle



$x_1^2 - x_2^2$: a hyperbola



$(x_1 + x_2)^2$: two lines

a direction with	**meaning**	**examples from Part II**
$Q(\mathbf{v}) > 0$	positive squared length	any Euclidean or Mahalanobis offset
$Q(\mathbf{v}) < 0$	negative squared length	the Lorentz axis x_0
$Q(\mathbf{v}) = 0$, $\mathbf{v} \neq \mathbf{0}$	null: nonzero, length zero	a constant labeling; $(1, 1, 0)$; the head's $(1, -1)$

Sylvester: Only the Count of Signs Survives

Any symmetric S can be brought, by a change of basis, to a diagonal matrix with entries $+1$, -1, and 0 only. **Sylvester's law of inertia**: the counts (p, q, r) of each do not depend on the basis chosen. They are the **signature** of the form, written $\mathrm{Cl}(p, q, r)$.

quadratic form	**eigenvalues of S**	**signature**
squared Euclidean length on $\mathbb{R}^n$	all 1	$\mathrm{Cl}(n, 0, 0)$
Mahalanobis, Σ^{-1} above	$0.56,\ 5.00$	$\mathrm{Cl}(2, 0, 0)$
$x_1^2 + 4x_1x_2 + x_2^2$	$3,\ -1$	$\mathrm{Cl}(1, 1, 0)$
x_1x_2	$\frac{1}{2},\ -\frac{1}{2}$	$\mathrm{Cl}(1, 1, 0)$
Lorentz, $-x_0^2 + x_1^2 + x_2^2$	$1,\ 1,\ -1$	$\mathrm{Cl}(2, 1, 0)$
Laplacian of the five-word chain	$0,\ 0.38,\ 1.38,\ 2.62,\ 3.62$	$\mathrm{Cl}(4, 0, 1)$
the head, $(x_1 + x_2)^2$	$2,\ 0$	$\mathrm{Cl}(1, 0, 1)$
a tied head, $W_Q^\top W_Q$ of rank k	k positive, the rest 0	$\mathrm{Cl}(k, 0, n-k)$

The signature is everything about a quadratic form that no change of coordinates can alter. Session 10 introduced $\mathrm{Cl}(p, q, r)$ as a name for a form; Part III shows why the letters are Cl.

The Head's Quadratic Form Is Degenerate

The head's $S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has eigenvalues 2 and 0: signature $\mathrm{Cl}(1,0,1)$, with null direction $(1,-1)$. Now look at the two tokens:

$$\mathbf{x}_1 - \mathbf{x}_2 = (2,\,-2), \qquad Q(\mathbf{x}_1 - \mathbf{x}_2) = (2-2)^2 = 0$$

The two tokens differ by a null vector. Under the symmetric half they cannot be told apart by any comparison at all:

$$S(\mathbf{x}_1, \mathbf{y}) \;=\; 4\,(y_1 + y_2) \;=\; S(\mathbf{x}_2, \mathbf{y}) \;\text{ for every } \mathbf{y}, \qquad \cos_S(\mathbf{x}_1, \mathbf{x}_2) = \frac{16}{\sqrt{16 \cdot 16}} = 1$$

Everything that distinguishes token 1 from token 2 for this head lives in A – exactly the half its quadratic form cannot see.

Why this matters

Session 11 found that a head with tied weights, $W_Q = W_K$, cannot break a tie between two equally similar tokens. The quadratic form shows how complete the tie is: for the symmetric half, these two tokens are one direction.

In-Class Exercise 1

1. For $G = \left(\begin{smallmatrix} 2 & 3 \\ -1 & 1 \end{smallmatrix}\right)$, write $Q(\mathbf{x}) = \mathbf{x}^{\top} G \mathbf{x}$ as a polynomial, and give the symmetric matrix that represents it.
2. Using *only* values of that Q, compute $S\big((1,0),(0,1)\big)$ by polarization. Check it against your matrix.
3. Find the signature of $Q(\mathbf{x}) = x_1 x_2$ and two null directions.
4. $\star$ On the chain good–decent–okay–mediocre–bad, why can the energy $\mathbf{x}^{\top} L \mathbf{x}$ alone never decide whether okay is positive or negative? What do the seeds change, in terms of the signature?

Six minutes. Item 4 is the starred one.

In-Class Exercise 1: Answers

1. $Q = 2x_1^2 + (3-1)\,x_1x_2 + x_2^2 = 2x_1^2 + 2x_1x_2 + x_2^2$, so $S = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$. The 3 and the -1 are averaged; their difference, the antisymmetric part, is gone.

2. $Q(1,1) = 5$, $Q(1,0) = 2$, $Q(0,1) = 1$, so $S = \frac{1}{2}(5-2-1) = 1$, the off-diagonal entry of the matrix.

3. $S = \left(\begin{smallmatrix} 0 & 1/2 \\ 1/2 & 0 \end{smallmatrix}\right)$ has eigenvalues $\frac{1}{2}$ and $-\frac{1}{2}$: signature $\mathrm{Cl}(1,1,0)$. The axes are null, $Q(1,0) = Q(0,1) = 0$, while $Q(1,1) = 1$ and $Q(1,-1) = -1$.

4. $\star$ Adding the same constant to every label leaves every difference $x_i - x_j$ unchanged, so the energy cannot see the level: the constant vector is a null direction, and the chain's signature is $\mathrm{Cl}(4,0,1)$. The seeds fix two values, which removes that null direction; the remaining problem has a unique minimum, with okay at 0.

Part III
From a Quadratic Form to a Product

Declare Q, Then Ask for a Product

Session 11 asked for a product whose two halves were the dot product and the wedge. Ask instead for something that mentions only the quadratic form:

The specification

Find an associative product of vectors, distributive over addition, such that **every vector squares to its quadratic form**:

$$\mathbf{v}\,\mathbf{v} \;=\; Q(\mathbf{v}) \qquad \text{for every vector } \mathbf{v}$$

That single rule is the whole definition of the **Clifford algebra** $\mathrm{Cl}(V, Q)$, which Clifford introduced in 1878 by combining Grassmann's algebra of the wedge with Hamilton's quaternions.

Notice what is *not* in the specification: no antisymmetric form, no wedge, no orientation. Q is the specification, and everything else is derived from it. Session 13b takes this literally: Q is the interpreter, and products of vectors are the programs it runs.

The left side is a product of vectors; the right side is a number. A product of two vectors is therefore allowed to be a number plus something else – and the next two slides find out what else.

Polarization, Inside the Product

Apply the rule to a sum, $\mathbf{u}+\mathbf{v}$, and expand by distributivity, keeping the order of each product:

$$(\mathbf{u}+\mathbf{v})(\mathbf{u}+\mathbf{v}) \;=\; \mathbf{uu}+\mathbf{uv}+\mathbf{vu}+\mathbf{vv} \;=\; Q(\mathbf{u}+\mathbf{v})$$

Replace $\mathbf{uu}$ and $\mathbf{vv}$ by $Q(\mathbf{u})$ and $Q(\mathbf{v})$, and solve:

$$\mathbf{uv}+\mathbf{vu} \;=\; Q(\mathbf{u}+\mathbf{v})-Q(\mathbf{u})-Q(\mathbf{v}) \;=\; 2\,S(\mathbf{u},\mathbf{v})$$

The symmetric half of the product is forced: it is the polarization of the declared Q. With the head's form and its two tokens, $(\mathbf{x}_1+\mathbf{x}_2)^2=64$ and $\mathbf{x}_1^2=\mathbf{x}_2^2=16$, so $\mathbf{x}_1\mathbf{x}_2+\mathbf{x}_2\mathbf{x}_1=32$.

The polarization identity of Part I has reappeared inside the product. Squaring a sum produces both orders of the cross term, and adding the two orders is how a quadratic form returns its symmetric half.

The Other Half Is Left Free

The rule says nothing about the difference $\mathbf{uv} - \mathbf{vu}$. The algebra keeps it, as an object of a new kind, the **wedge**:

$$\mathbf{u} \wedge \mathbf{v} \;\coloneqq\; \tfrac{1}{2}\,(\mathbf{uv} - \mathbf{vu}), \qquad \boxed{\mathbf{uv} \;=\; S(\mathbf{u},\mathbf{v}) \;+\; \mathbf{u} \wedge \mathbf{v}}$$

half	**determined by**	**grade**	**answers**
symmetric, $S(\mathbf{u},\mathbf{v})$	the declared Q, by polarization	0, a number	how alike
antisymmetric, $\mathbf{u} \wedge \mathbf{v}$	the two vectors alone: no metric needed	2, a bivector	which way

Session 11 derived $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ for the ordinary dot product. Nothing in the derivation above used the dot product: the same formula holds for every quadratic form, with S in place of the dot.

This answers the puzzle. The declaration fixes only Q. The antisymmetric half was never declared, because it never needed declaring: the wedge of two vectors does not depend on Q at all.

The Head's Two Halves, Recomputed

Declare the head's quadratic form, $Q(\mathbf{x}) = (x_1 + x_2)^2$, and multiply the two tokens in its Clifford algebra:

$$\mathbf{x}_1\mathbf{x}_2 = \underbrace{16}_{S(\mathbf{x}_1,\mathbf{x}_2)} + \underbrace{8\,\mathbf{e}_1 \wedge \mathbf{e}_2}_{\mathbf{x}_1 \wedge \mathbf{x}_2}, \qquad \mathbf{x}_2\mathbf{x}_1 = 16 - 8\,\mathbf{e}_1 \wedge \mathbf{e}_2$$

since $\mathbf{x}_1 \wedge \mathbf{x}_2 = (3 \cdot 3 - 1 \cdot 1)\,\mathbf{e}_1 \wedge \mathbf{e}_2$. Set that beside what the head computed:

	the head's score	**the Clifford product**
symmetric half, $1 \to 2$ and $2 \to 1$	$16,\ 16$	$16,\ 16$ (declared, via Q)
antisymmetric half, $1 \to 2$ and $2 \to 1$	$+8,\ -8$	$+8,\ -8$ (computed, not declared)

The head had to learn both halves. The algebra was told only the first. The second agrees because in the plane every antisymmetric form is a multiple of the wedge, and this head happened to learn the multiple 1.

Under this Q, $\mathbf{e}_1$ and $\mathbf{e}_2$ are not orthogonal, so $\mathbf{e}_1\mathbf{e}_2 \neq \mathbf{e}_1 \wedge \mathbf{e}_2$; the wedge is written out explicitly for that reason.

A Basis Gives the Whole Multiplication Table

Choose a basis that is orthogonal for S. Polarization in the product gives the two rules that generate everything:

$$\mathbf{e}_i\mathbf{e}_j = -\,\mathbf{e}_j\mathbf{e}_i \quad (i \neq j), \qquad\qquad \mathbf{e}_i\mathbf{e}_i = Q(\mathbf{e}_i)$$

By Sylvester, the basis can be scaled so that each $Q(\mathbf{e}_i)$ is $+1$, -1, or 0. With p, q, and r of each, the algebra is $\mathrm{Cl}(p, q, r)$ – the name Session 10 gave the form, now the name of the algebra it generates.

The head's algebra. $\mathbf{g}_1 = (\frac{1}{2}, \frac{1}{2})$ and $\mathbf{g}_2 = (1, -1)$ are orthogonal for S, with $Q(\mathbf{g}_1) = 1$ and $Q(\mathbf{g}_2) = 0$: the algebra is $\mathrm{Cl}(1, 0, 1)$.

$\mathbf{x}_1 = 4\mathbf{g}_1 + \mathbf{g}_2, \quad \mathbf{x}_2 = 4\mathbf{g}_1 - \mathbf{g}_2$	coordinates in the new basis
$\mathbf{x}_1\mathbf{x}_2 = 16\,\mathbf{g}_1\mathbf{g}_1 - 4\,\mathbf{g}_1\mathbf{g}_2 + 4\,\mathbf{g}_2\mathbf{g}_1 - \mathbf{g}_2\mathbf{g}_2 = 16 - 8\,\mathbf{g}_1\mathbf{g}_2$	the same product, from the table
$(\mathbf{x}_1 - \mathbf{x}_2)^2 = (2\mathbf{g}_2)^2 = 0$	the null difference again

And $\mathbf{g}_1\mathbf{g}_2 = \mathbf{g}_1 \wedge \mathbf{g}_2 = -\,\mathbf{e}_1 \wedge \mathbf{e}_2$, so $-8\,\mathbf{g}_1\mathbf{g}_2$ is the $+8\,\mathbf{e}_1 \wedge \mathbf{e}_2$ of the last slide.

What the Tensor Algebra Would Have Kept

The outer product never forgets: $\mathbf{e}_2 \otimes \mathbf{e}_1$, $\mathbf{e}_2 \otimes \mathbf{e}_1 \otimes \mathbf{e}_2$, and so on are all independent, and products of degree k need n^k coordinates – the order growth of Session 11b, and the cost of TPR.

The Clifford rule cuts this down. Every word in the basis vectors reduces by two moves: swap neighbors (a sign, $\mathbf{e}_j\mathbf{e}_i = -\mathbf{e}_i\mathbf{e}_j$) and cancel equal neighbors ($\mathbf{e}_i\mathbf{e}_i = Q(\mathbf{e}_i)$). Take the word $\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2\mathbf{e}_1$:

declared signature		reduction	value
$\mathrm{Cl}(2,0,0)$:	$\mathbf{e}_1^2 = \mathbf{e}_2^2 = +1$	$\mathbf{e}_2\mathbf{e}_1\,(\mathbf{e}_2\mathbf{e}_2)\,\mathbf{e}_1 = \mathbf{e}_2\,(\mathbf{e}_1\mathbf{e}_1)$	$\mathbf{e}_2$
$\mathrm{Cl}(1,1,0)$:	$\mathbf{e}_2^2 = -1$	$\mathbf{e}_2\mathbf{e}_1\,(-1)\,\mathbf{e}_1 = -\,\mathbf{e}_2\,(\mathbf{e}_1\mathbf{e}_1)$	$-\mathbf{e}_2$
$\mathrm{Cl}(1,0,1)$:	$\mathbf{e}_2^2 = 0$	$\mathbf{e}_2\mathbf{e}_1\,(0)\,\mathbf{e}_1$	0

One word, three values. The tensor algebra cannot evaluate it; the Clifford algebra can, and what it returns depends on nothing but the declared quadratic form. Session 13b starts from this slide: the two moves are the interpreter, and a word is a program.

Formally, $\mathrm{Cl}(V, Q)$ is the tensor algebra with $\mathbf{v} \otimes \mathbf{v} - Q(\mathbf{v})$ set to zero for every $\mathbf{v}$. Every word reduces to $\pm$ a sorted product of *distinct* basis vectors, or to 0.

Tensors Grow; the Clifford Algebra Stays Put

A reduced word is a choice of *which* basis vectors appear: 2^n of them, sorted by **grade**, the number of vectors in each.

$n = 4$, degree or grade k	0	1	2	3	4	**total**
tensor products, n^k	1	4	16	64	256	keeps growing
Clifford grade k, $\binom{n}{k}$	1	4	6	4	1	$16 = 2^4$

A product of two vectors lands in grades 0 and 2: the symmetric half in grade 0, the antisymmetric half in grade 2, each in a place of its own. At $n = 300$, grades 0 through 2 hold $1 + 300 + 44{,}850 = 45{,}151$ coordinates, against $90{,}000$ for a 300×300 matrix. The saving has a price: grade 2 is alternating, so it drops the symmetric and diagonal structure that a general relation needs (13c).

	size	**antisymmetric half**	**unbinding**
TPR	grows as n^k	kept	exact, with orthonormal roles
VSA (HRR, MAP)	fixed, d	folded away	approximate; needs cleanup
Clifford algebra	fixed, 2^n, graded	kept, in grade 2	exact, with orthonormal roles

Part IV
The Smallest Algebras

what one or two declared letters generate

The Coefficients Stay Real; the New Letter Is a Direction

One distinction first. The symbol $\mathbb{R}$ plays two different parts:

As the scalars

Numbers such as 2, -3.7, π are **coefficients**. They say how much of a vector there is, and they multiply as numbers always did.

As a one-dimensional vector space

$V = \mathbb{R}\,\mathbf{e}$ has vectors $\mathbf{v} = x\,\mathbf{e}$ with $x \in \mathbb{R}$. Here $\mathbf{e}$ is a **basis direction**, a letter of the alphabet, not a scalar.

The coefficients stay real in everything that follows. The only new question is what $\mathbf{ee}$ should be, and the declaration answers it: $\mathbf{ee} = Q(\mathbf{e})$, with $Q(x\,\mathbf{e}) = Q(\mathbf{e})\,x^2$.

declaration	**quadratic form**	**so**
$\mathbf{e}^2 = +1$	$Q(x\,\mathbf{e}) = +x^2$	$(3\mathbf{e})^2 = 9$
$\mathbf{e}^2 = -1$	$Q(x\,\mathbf{e}) = -x^2$	$(3\mathbf{e})^2 = -9$
$\mathbf{e}^2 = 0$	$Q(x\,\mathbf{e}) = 0$	$(3\mathbf{e})^2 = 0$

A general vector need not square to $+1$, -1 or 0; those are the normalized values for basis letters, as Sylvester's theorem allows.

One Generator, Three Algebras

Start with a single basis vector $\mathbf{e}$. The algebra has elements $a + b\,\mathbf{e}$, and the declared sign of $\mathbf{e}^2$ is the only choice there is:

$$(a + b\,\mathbf{e})(c + d\,\mathbf{e}) \;=\; \big(ac + bd\,\mathbf{e}^2\big) + (ad + bc)\,\mathbf{e}$$

$\mathbf{e}^2$	**algebra**	**worked**	**known as**
$+1$	$\mathrm{Cl}(1,0,0)$	$(2+3\mathbf{e})(1+4\mathbf{e}) = 14 + 11\mathbf{e}$	split-complex numbers
-1	$\mathrm{Cl}(0,1,0)$	$(2+3\mathbf{e})(1+4\mathbf{e}) = -10 + 11\mathbf{e}$	the complex numbers
0	$\mathrm{Cl}(0,0,1)$	$(3+\mathbf{e})(3+\mathbf{e}) = 9 + 6\mathbf{e}$	dual numbers

One sign, three different worlds. Session 13b shows what each world is good for: $+1$ splits things into two exclusive pieces, -1 turns them in a cycle, and 0 carries a rate of change.

The complex numbers are not postulated here. They are the Clifford algebra of a single direction declared to have squared length -1.

Squaring in Each Algebra

Keep the multiplication algorithm fixed, $(a + b\,\mathbf{e})(c + d\,\mathbf{e}) = (ac + bd\,\mathbf{e}^2) + (ad + bc)\,\mathbf{e}$, and change only the line that says what $\mathbf{e}^2$ is. Square $2 + 3\mathbf{e}$ three times:

declaration	**expansion**	**result**	**known as**
$\mathbf{e}^2 = -1$, write $\mathbf{e} = i$	$4 + 12i + 9i^2 = 4 + 12i - 9$	$-5 + 12i$	complex numbers
$\mathbf{e}^2 = +1$, write $\mathbf{e} = j$	$4 + 12j + 9j^2 = 4 + 12j + 9$	$13 + 12j$	split-complex numbers
$\mathbf{e}^2 = 0$, write $\mathbf{e} = \varepsilon$	$4 + 12\varepsilon + 9\varepsilon^2 = 4 + 12\varepsilon + 0$	$4 + 12\varepsilon$	dual numbers

Only the $9\,\mathbf{e}^2$ term differs, and it differs by nothing but the declared sign. Same coefficients, different squaring rule, different algebra.

In the complex case there is no mysterious operation: the real scalars were enlarged by one direction, and that direction was told what its square is. The -9 is the declaration showing up inside a computation.

The dual-number row is the one in which the second-order term vanishes. Session 13b shows that this is what makes ε carry a derivative.

Two Generators: The Sign of the Bivector

With two orthogonal basis vectors, the bivector $\mathbf{e}_1\mathbf{e}_2$ appears, and its square is decided by the declared signs:

$$(\mathbf{e}_1\mathbf{e}_2)^2 \;=\; \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2 \;=\; -\,\mathbf{e}_1\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2 \;=\; -\,Q(\mathbf{e}_1)\,Q(\mathbf{e}_2)$$

signature	$\mathbf{e}_1^2,\ \mathbf{e}_2^2$	$(\mathbf{e}_1\mathbf{e}_2)^2$	**what lives there**
$\mathrm{Cl}(2,0,0)$	$+1,\ +1$	-1	rotations of the plane; the complex numbers inside it
$\mathrm{Cl}(1,1,0)$	$+1,\ -1$	$+1$	boosts, as in special relativity (Session 13b)
$\mathrm{Cl}(0,2,0)$	$-1,\ -1$	-1	the quaternions
$\mathrm{Cl}(1,0,1)$	$+1,\ 0$	0	translations (Session 13b)

Quaternions, for free. In $\mathrm{Cl}(0,2,0)$ put $i = \mathbf{e}_1$, $j = \mathbf{e}_2$, $k = \mathbf{e}_1\mathbf{e}_2$. Then $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, and $ijk = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2 = -\mathbf{e}_1\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2 = -(-1)(-1) = -1$: Hamilton's rules of 1843 are the multiplication table of a declared quadratic form with two minus signs. Session 14 finds the same rules among the planes of ordinary space.

A Small Family of Declarations

letters declared	squares	algebra	known as	dim.
none	–	$\mathrm{Cl}(0,0,0)$	the real numbers	1
one negative	$\mathbf{e}^2 = -1$	$\mathrm{Cl}(0,1,0)$	the complex numbers	2
one positive	$\mathbf{e}^2 = +1$	$\mathrm{Cl}(1,0,0)$	split-complex numbers	2
one null	$\mathbf{e}^2 = 0$	$\mathrm{Cl}(0,0,1)$	dual numbers	2
two negative	$\mathbf{e}_1^2 = \mathbf{e}_2^2 = -1$	$\mathrm{Cl}(0,2,0)$	the quaternions	4
two positive	$\mathbf{e}_1^2 = \mathbf{e}_2^2 = +1$	$\mathrm{Cl}(2,0,0)$	the plane; even part: the complex numbers	4
three positive	all $+1$	$\mathrm{Cl}(3,0,0)$	space; even part: the quaternions (Session 14)	8

Two routes to the complex numbers: one letter declared -1, or the even programs $a + b\,\mathbf{e}_1\mathbf{e}_2$ of the Euclidean plane, where the bivector plays i. In the second, i is an oriented plane element, not an added scalar direction. That is the reading Session 13b develops.

The quadratic form does not merely report geometry. Once imposed as $\mathbf{v}^2 = Q(\mathbf{v})$, it fixes the multiplication table of the entire algebra, and with it which familiar number system the algebra is.

In-Class Exercise 2

1. In $\mathrm{Cl}(2,0,0)$, compute $(\mathbf{e}_1 + 2\mathbf{e}_2)(3\mathbf{e}_1 - \mathbf{e}_2)$ by the two moves, and split the result into its symmetric and antisymmetric halves.
2. Repeat the computation in $\mathrm{Cl}(1,1,0)$, where $\mathbf{e}_2^2 = -1$. Which half changed, and why that one?
3. Multiply $(2 + 3\mathbf{e})(1 + 4\mathbf{e})$ under each of the three one-letter declarations.
4. $\star$ In $\mathrm{Cl}(2,0,0)$ the bivector $\mathbf{e}_1\mathbf{e}_2$ squares to -1, and in $\mathrm{Cl}(0,2,0)$ so does the letter $\mathbf{e}_1$. Both algebras contain a copy of the complex numbers. What distinguishes them?

Seven minutes. Item 2 is the one to discuss.

In-Class Exercise 2: Answers

1. $3\mathbf{e}_1\mathbf{e}_1 - \mathbf{e}_1\mathbf{e}_2 + 6\mathbf{e}_2\mathbf{e}_1 - 2\mathbf{e}_2\mathbf{e}_2 = 3 - \mathbf{e}_1\mathbf{e}_2 - 6\mathbf{e}_1\mathbf{e}_2 - 2 = 1 - 7\,\mathbf{e}_1\mathbf{e}_2$. Symmetric half 1 (the dot product, $3-2$); antisymmetric half $-7\,\mathbf{e}_1\mathbf{e}_2$ (the wedge, $1\cdot(-1) - 2\cdot 3$).

2. Now $\mathbf{e}_2\mathbf{e}_2 = -1$, so the result is $3 + 2 - 7\,\mathbf{e}_1\mathbf{e}_2 = 5 - 7\,\mathbf{e}_1\mathbf{e}_2$. Only the symmetric half changed: it is the polarization of Q, and Q is what was changed. The wedge does not depend on Q, so it stayed at -7.

3. $(ac + bd\,\mathbf{e}^2) + (ad + bc)\,\mathbf{e}$ with $a, b, c, d = 2, 3, 1, 4$: $\mathbf{e}^2 = +1$ gives $14 + 11\mathbf{e}$; $\mathbf{e}^2 = -1$ gives $-10 + 11\mathbf{e}$; $\mathbf{e}^2 = 0$ gives $2 + 11\mathbf{e}$. The $\mathbf{e}$ coefficient never sees the declaration.

4. ⋆ The copies of the complex numbers agree: $a + b\,\mathbf{e}_1\mathbf{e}_2$ and $a + b\,\mathbf{e}_1$ multiply alike. The algebras differ in their *vectors*: in $\mathrm{Cl}(2,0,0)$ the letters square to $+1$ and can be mirrors (Session 13b), while in $\mathrm{Cl}(0,2,0)$ they square to -1 and are themselves imaginary units, i and j. One algebra is four numbers holding a plane; the other is four numbers holding the quaternions.

Where Each Ingredient Comes From: The Next Two Sessions

declared, today	**computed by the algebra**	**added by FGA**	**where**
the quadratic form Q	S by polarization: the symmetric half	a signature chosen for meaning	13b
–	$\mathbf{u} \wedge \mathbf{v}$: the antisymmetric half	subspaces for roles and slots	13c
an orthogonal basis	$\mathrm{Cl}(p, q, r)$: the whole table	grade and subspace as type	13c
–	inverses, mirrors, rotors	wedge binding; exact unbinding	13b, 13c
–	projectors, cycles, rates	polar pairs; coercion by rotor	13b

Read the first column: the only thing declared is a quadratic form. Everything in the second column follows from $\mathbf{v}^2 = Q(\mathbf{v})$: today derived the first three rows, and 13b derives the rest. The third column is what Functional Geometric Algebra builds on top, in 13b and 13c.

Session 10 asked what the matrix in the middle of $\mathbf{x}^\top G \mathbf{y}$ should be, and Session 11 split it in two. The Clifford algebra is the answer that declares one half and derives the other.

Summary

1. A **quadratic form** is a bilinear form with the same vector in both slots, $Q(\mathbf{x}) = \mathbf{x}^\top G\mathbf{x}$. Since $\mathbf{x}^\top A\mathbf{x} = 0$, it sees only the symmetric half.
2. **Polarization**, $S(\mathbf{x},\mathbf{y}) = \frac{1}{2}[Q(\mathbf{x}+\mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y})]$, recovers that half exactly. A quadratic form and a symmetric bilinear form are the same information.
3. Quadratic forms are everywhere in NLP: Mahalanobis distance for out-of-domain detection, the Laplacian energy of label propagation, the Lorentz form of hyperbolic embeddings.
4. By Sylvester, a quadratic form is classified by its **signature** (p, q, r). The attention head of Session 11 has signature $(1, 0, 1)$, and its two tokens differ by a null vector.
5. The **Clifford algebra** is defined by one rule, $\mathbf{v}^2 = Q(\mathbf{v})$. Polarization forces the symmetric half of the product; the antisymmetric half, the wedge, comes free: $\mathbf{uv} = S(\mathbf{u},\mathbf{v}) + \mathbf{u} \wedge \mathbf{v}$. Two moves, swap and cancel, evaluate every word, and one word can take three values under three declarations.
6. The **smallest algebras** are familiar number systems: one letter declared -1, $+1$ or 0 gives the complex, split-complex and dual numbers; two letters declared -1 give the quaternions. $i^2 = -1$ is a declaration, not a discovery.

Coming up: 13b reads Q as an interpreter and products as programs; 13c builds the wedge and the dual and returns to binding; Session 14 gives the full table of $\mathrm{Cl}(3, 0, 0)$.

Exercises

1. Prove that $\mathbf{x}^\top A\mathbf{x} = 0$ for every antisymmetric A, and conclude that no function of Q can recover the antisymmetric half of a form.
2. Show that polarizing $Q(\mathbf{x}) = \mathbf{x}^\top S\mathbf{x}$ returns $S(\mathbf{x}, \mathbf{y}) = \mathbf{x}^\top S\mathbf{y}$ for every symmetric S.
3. Find the signatures of the Laplacians of a triangle (three words, all linked) and of two separate edges (four words). Relate the number of zeros to the number of connected components.
4. In $\mathrm{Cl}(3,0,0)$, show that $(\mathbf{e}_1\mathbf{e}_2)^2 = (\mathbf{e}_2\mathbf{e}_3)^2 = (\mathbf{e}_3\mathbf{e}_1)^2 = -1$. Compute $(\mathbf{e}_2\mathbf{e}_3)(\mathbf{e}_3\mathbf{e}_1)$, and find signs for i, j, k that make these bivectors obey $ij = k$.
5. Reduce the word $\mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_3\mathbf{e}_2$ by the two moves in $\mathrm{Cl}(3,0,0)$, then in $\mathrm{Cl}(2,1,0)$ with $\mathbf{e}_3^2 = -1$, then in $\mathrm{Cl}(2,0,1)$ with $\mathbf{e}_3^2 = 0$.
6. $\star$ Show that in $\mathrm{Cl}(2,0,0)$ the four elements $1, \mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_1\mathbf{e}_2$ are closed under the product, and write out the full 4×4 table. Which entries would change in $\mathrm{Cl}(1,1,0)$, and which would not?

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